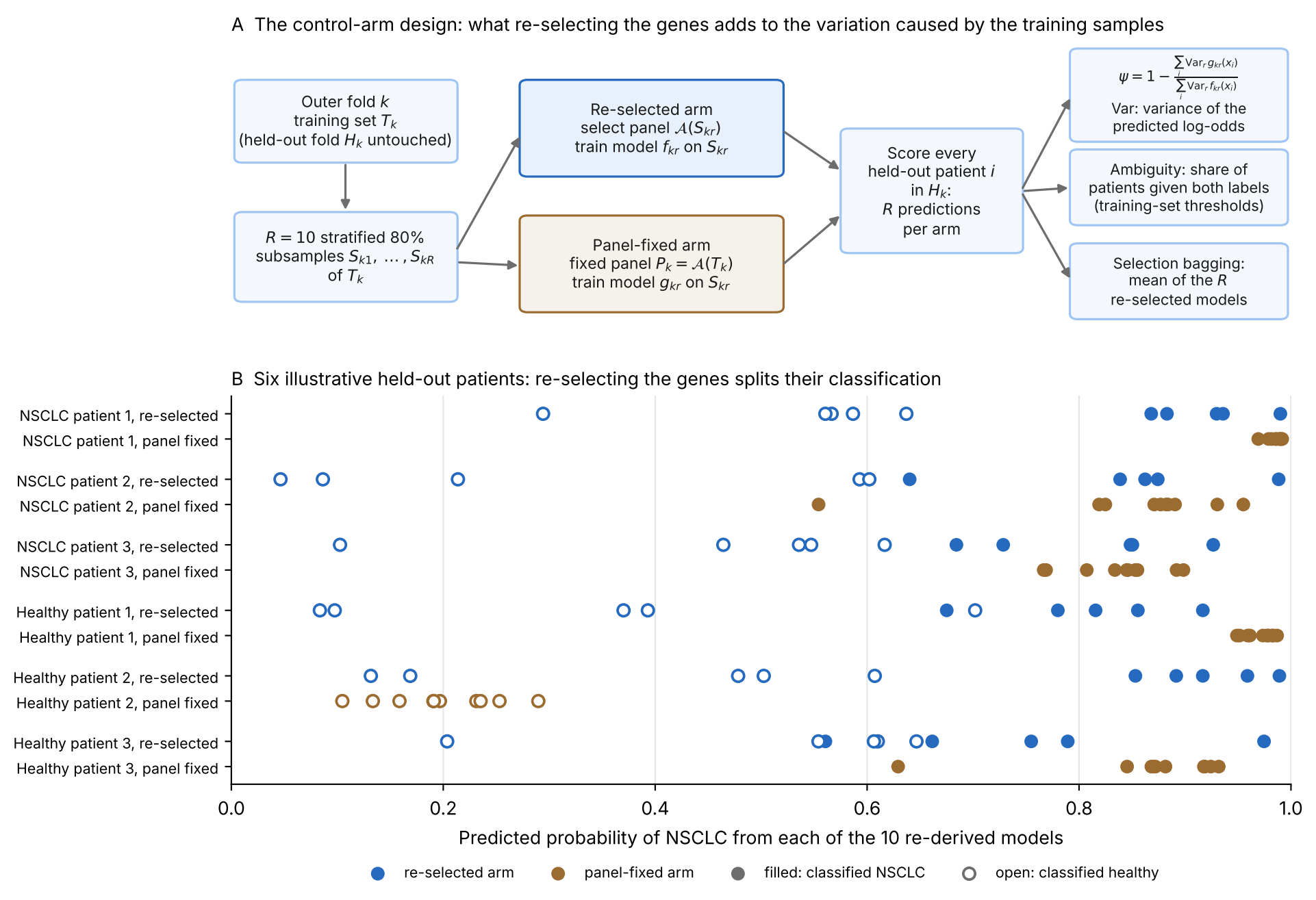
A The control-arm design: what re-selecting the genes adds to the variation caused by the training samples
Outer fold k training set T_k (held-out fold H_k untouched)
$R = 10$ stratified 80% subsamples $S_{k1}, \ldots, S_{kR}$ of T_k
Re-selected arm select panel $\mathcal{A}(S_{kr})$ train model f_{kr} on S_{kr}
Panel-fixed arm fixed panel $P_k = \mathcal{A}(T_k)$ train model g_{kr} on S_{kr}
Score every held-out patient i in H_k: R predictions per arm
$\psi = 1 - \frac{\sum_i \mathrm{Var}_r\, g_{kr}(x_i)}{\sum_i \mathrm{Var}_r\, f_{kr}(x_i)}$
Var: variance of the predicted log-odds
Ambiguity: share of patients given both labels (training-set thresholds)
Selection bagging: mean of the R re-selected models
B Six illustrative held-out patients: re-selecting the genes splits their classification
NSCLC patient 1, re-selected
NSCLC patient 1, panel fixed
NSCLC patient 2, re-selected
NSCLC patient 2, panel fixed
NSCLC patient 3, re-selected
NSCLC patient 3, panel fixed
Healthy patient 1, re-selected
Healthy patient 1, panel fixed
Healthy patient 2, re-selected
Healthy patient 2, panel fixed
Healthy patient 3, re-selected
Healthy patient 3, panel fixed
0.0
0.2
0.4
0.6
0.8
1.0
Predicted probability of NSCLC from each of the 10 re-derived models
re-selected arm
panel-fixed arm
filled: classified NSCLC
open: classified healthy